

QoL - Primary Estimand

Here we analyse a random sample of the simulated data as we would analyze the true data of the liqplat trial.

Setup

```
library(arrow)
library(tidyverse)
library(data.table)
library(rmsb)
library(ggplot2)
library(ggdist)
library(patchwork)
library(furrr)
library(here)
```

Load data

```
set.seed(12)
# Load full dataset
full_data <- read_parquet(here::here("1-qol", "sim-data", "sim_data_nonpo.parquet"))
generate_SOPS <- FALSE
fit <- FALSE
```

Sample and format data

```
# --- 2. Sample Patients ---
ss_tx <- 180
ss_control <- 90

tx_ids <- sample(full_data |> filter(tx == 1) |> pull(id) |> unique(), ss_tx, replace =
FALSE)
control_ids <- sample(full_data |> filter(tx == 0) |> pull(id) |> unique(), ss_control,
replace = FALSE)

df <- full_data |>
  filter(id %in% c(tx_ids, control_ids))

rm(full_data)

# --- 3. Pre-process Data & Save Empirical SOPs ---
ids_death <- df$id[df$y == '8']
df_filtered <- df |>
  filter(!(id %in% ids_death))

# Save empirical SOP image for this iteration's data
p_empirical <- df_filtered |>
  group_by(time, y, tx) |>
  summarise(
    n = n()
```

```

) |>
group_by(time, tx) |>
mutate(
  prop = n / sum(n),
) |>
ungroup() |>
mutate(
  y = factor(y),
  tx = factor(tx)) |>
ggplot(aes(x = time, y = prop)) +
  geom_line(aes(color = y, linetype = tx)) +
  scale_color_brewer(palette = "Dark2", guide="none") +
  scale_x_continuous(breaks = seq(1, 26, by=4)) +
  scale_y_continuous(breaks = seq(0, 1, by=0.1)) +
  coord_cartesian(ylim = c(0, 0.5)) +
  labs(x = "Study Week",
  y = "Proportion",
  subtitle = "Empirical state occupancy probabilities (Unkown)",
  fill = "State",
  linetype = "Treatment") +
  theme(
    legend.position = ""
  )

```

`summarise()` has grouped output by 'time', 'y'. You can override using the `.groups` argument.

```

# --- 4. Create analysis dataset with missing follow-up ---
baseline <- df_filtered |>
  filter(time == 1) |>
  mutate(y = yprev, time = 0) |>
  select(-yprev)

follow_up <- df_filtered |>
  slice_sample(prop = 0.15) |>
  select(-yprev)

data_for_model <- baseline |>
  bind_rows(follow_up) |>
  arrange(id, time) |>
  group_by(id) |>
  mutate(
    yprev = lag(y, n = 1, order_by = time),
    yprev = factor(yprev, levels = 1:7, ordered = FALSE),
    timeprev = lag(time, n = 1, order_by = time),
    gap = time - timeprev,
    y = as.numeric(as.character(y))
  ) |>
  ungroup() |>
  filter(!is.na(yprev)) |>
  select(-timeprev)

```

Visualize the follow-up pattern

```
set.seed(4)
ids <- data_for_model$id |> unique() |> sample(20)

fig_tile_random <- data_for_model |>
  filter(id %in% ids) |>
  mutate(id = as.factor(as.numeric(as.factor((id))))) |>
  ggplot(aes(y = id, x = time)) +
  geom_tile(mapping = aes(fill = factor(y)), width = 0.95, height = 0.4) +
  scale_fill_brewer(palette = "Dark2") +
  scale_x_continuous(breaks = 1:26) +
  labs(fill = "y")

ggsave(filename = here::here("img", "fig-sparse-tile.svg"), width = 12, height = 6)
```

Fit the model

```
if(fit){
  dd <- datadist(data_for_model)
  # Set the global option so that rms/rmsb functions know to use 'dd'
  options(datadist = 'dd')
  options(mc.cores = 8)

  Sys.setenv(RSTUDIO = 1)

  pcontrast_arg <- list(
    c1 = list(tx = 1),
    c2 = list(tx = 0),
    contrast = expression(c1 - c2),
    mu = 0,
    sd = 0.940
  )

  model <-
    blrm(
      formula = y ~ tx + rcs(time, 4) + yprev * gap + ecog_fstcnt + diagnosis,
      pcontrast = pcontrast_arg,
      data = data_for_model,
      ppo = ~ time,
      cppo = function(y) y,
      refresh = 5,
      iter = 2000,
      chains = 4,
      method = "sampling"
    )

  Sys.setenv(RSTUDIO = 0)

  saveRDS(model, here::here("1-qol", "models", "model-primary-estimand.rds"))} else{
  model <- readRDS(here::here("1-qol", "models", "model-primary-estimand.rds"))
}
```

Recovering difference in weeks with $\geq$ good QoL

Marginalized State Occupancy Probabilities

```
baseline_df <- data_for_model |>
  group_by(id) |>
  arrange(time) |>
  slice_head(n = 1) |>
  ungroup() |>
  mutate(
    time = 1,
    gap = 1,
    y = NA) |>
  as.data.table()
```

```
write_SOP <- function(i, tx, baseline_df, path){

  row <- baseline_df[i]

  sops <-
  soprobMarkov0rdm(
    model,
    data = list(tx = tx, ecog_fstcnt = row$ecog_fstcnt, diagnosis = row$diagnosis,
yprev=row$yprev, gap = row$gap),
    times = 1:26,
    ylevels = 1:7,
    absorb = NULL,
    tvarname = "time",
    pvarname = "yprev"
  )

  # Because of the size of the data, only use the first 500 MCMC draws
  sops <- sops[1:2000,,]

  sops <- as.data.table(sops)

  # Rename columns
  setnames(sops,
    old = c("V1", "V2", "V3", "value"),
    new = c("draw", "time", "state", "sop"))

  sops$time <- as.integer(sops$time)
  sops$state <- as.factor(sops$state)
  sops$tx <- tx
  sops$i <- i

  folder <- file.path(path, glue::glue("marginalized_sop_{tx}"),
glue::glue("msop_{i}.parquet"))

  # Create the directory if it doesn't exist
  dir_to_create <- dirname(folder)
  if (!dir.exists(dir_to_create)) {
    dir.create(dir_to_create, recursive = TRUE)
  }
}
```

```
arrow::write_parquet(  
  x = sops,  
  sink = folder  
)  
}
```

Generate marginalized SOPs

```
path <- here::here("1-qol", "output", "primary-estimand")
future::plan(future::multisession, workers=6)

# Control
if (generate_SOPS) {
  furrr::future_walk(1:nrow(baseline_df), \(x) write_SOP(x, 0, baseline_df, path),
    .progress=TRUE,
    .options = furrr_options(seed=TRUE))
}
```

```
# Treatment
if (generate_SOPS) {
  furrr::future_walk(1:nrow(baseline_df), \(x) write_SOP(x, 1, baseline_df, path),
    .progress=TRUE,
    .options = furrr_options(seed=TRUE))
}
```

Read maginalized SOPs

```
# Get list of all files
files <- fs::dir_ls(here::here("1-qol", "output", "primary-estimand",
"marginalized_sop_0"), glob = "*.parquet")

# Read in each parquet file as a data.frame and merge into a data.table
soc_df <- map(files, \(x) arrow::read_parquet(x), .progress=TRUE) |> rbindlist()
```

79% | ETA: 1s

```
# Average SOPs over covariate settings -- for each state, day, and MCMC draw
soc_df <- soc_df[, .(sop = mean(sop)), by = .(state, time, draw, tx)]
soc_df[, state := as.factor(state)]
```

```
# Get list of all files
files <- fs::dir_ls(here::here("1-qol", "output", "primary-estimand",
"marginalized_sop_1"), glob = "*.parquet")

# Read in each parquet file as a data.frame and merge into a data.table
tx_df <- map(files, \(x) arrow::read_parquet(x), .progress=TRUE) |> rbindlist()
```

91% | ETA: 0s

```
# Average SOPs over covariate settings -- for each state, day, and MCMC draw
tx_df <- tx_df[, .(sop = mean(sop)), by = .(state, time, draw, tx)]
tx_df[, state := as.factor(state)]
```

```
sop_df <-
  bind_rows(soc_df, tx_df) |>
  mutate(tx = as.factor(tx))
```

Plotting

Marginalized SOPs

```
fig_marginalized <-
sop_df |>
  ggplot() +
  aes(x = time, y = sop) +
  ggdist::stat_lineribbon(aes(fill = state, linetype = tx),
    linewidth = 0.6,
    alpha = 0.5,
    .width = c(0.95)) +
  scale_color_brewer(palette = "Dark2", guide="none") +
  scale_fill_brewer(palette = "Dark2") +
  scale_x_continuous(breaks = seq(1, 26, by=4)) +
  scale_y_continuous(breaks = seq(0, 1, by=0.1)) +
  coord_cartesian(ylim = c(0, 0.5)) +
  labs(x = "Study Week",
    y = "SOP",
    subtitle = "Marginalized state occupancy probabilities",
    fill = "State",
    linetype = "Treatment")

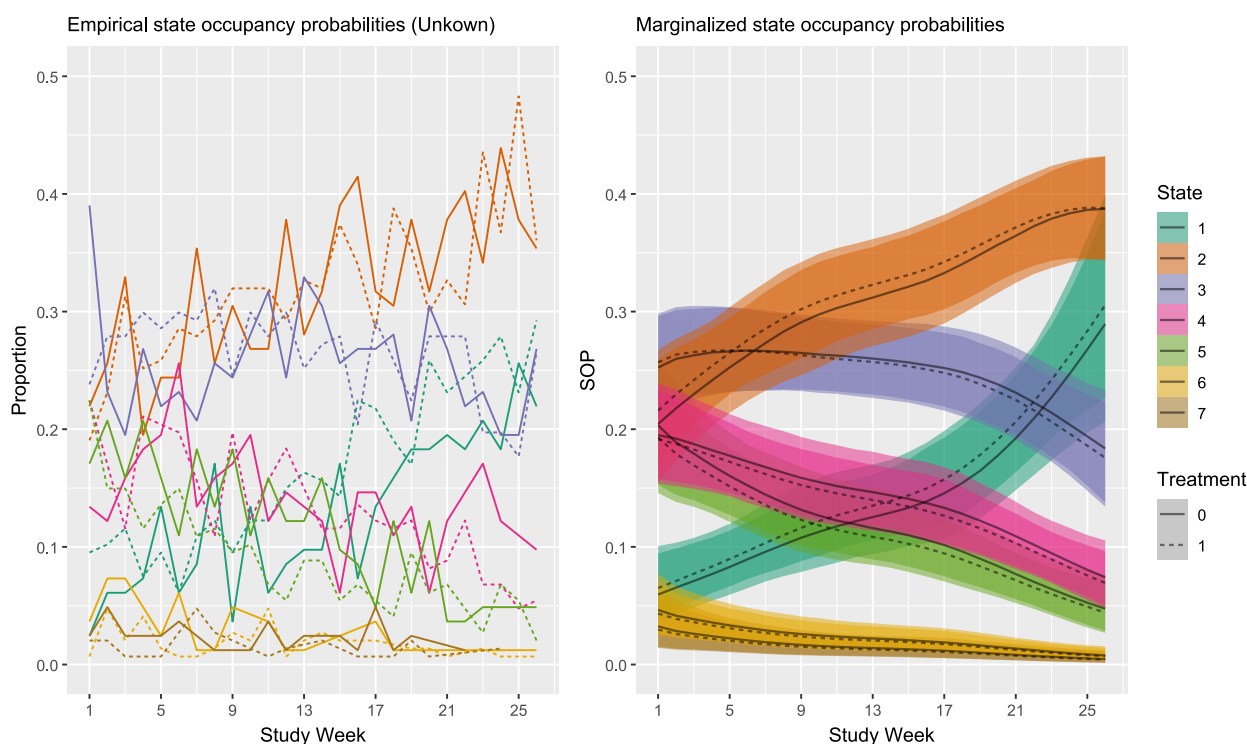
ggsave(plot = fig_marginalized, here::here("img", "primary-estimand-sop-line.svg"),
  width = 10, height = 6)

fig_marginalized_grouped <- sop_df |>
  mutate(group = case_when(
    state == 1 | state == 2 | state == 3 ~ "Good",
    state == 7 | state == 6 | state == 5 ~ "Bad",
    TRUE ~ "Average"
  )) |>
  group_by(group, draw, tx, time) |>
  summarise(sop = sum(sop)) |>
  ggplot() +
  aes(x = time, y = sop) +
  ggdist::stat_lineribbon(aes(fill = group, linetype = tx),
    linewidth = 0.6,
    alpha = 0.5,
    .width = c(0.95)) +
  scale_color_brewer(palette = "Dark2", guide="none") +
  scale_fill_brewer(palette = "Dark2") +
  scale_x_continuous(breaks = seq(1, 26, by=4)) +
  scale_y_continuous(breaks = seq(0, 1, by=0.1)) +
  coord_cartesian(ylim = c(0, 1)) +
```

```
labs(x = "Study Week",
     y = "SOP",
     subtitle = "Marginalized state occupancy probabilities",
     fill = "State",
     linetype = "Treatment")
```

``summarise()`` has grouped output by 'group', 'draw', 'tx'. You can override using the ``groups`` argument.

```
p_empirical + fig_marginalized + plot_layout(guides = "collect")
```



Difference in Weeks with $\geq$ Good Quality of Life

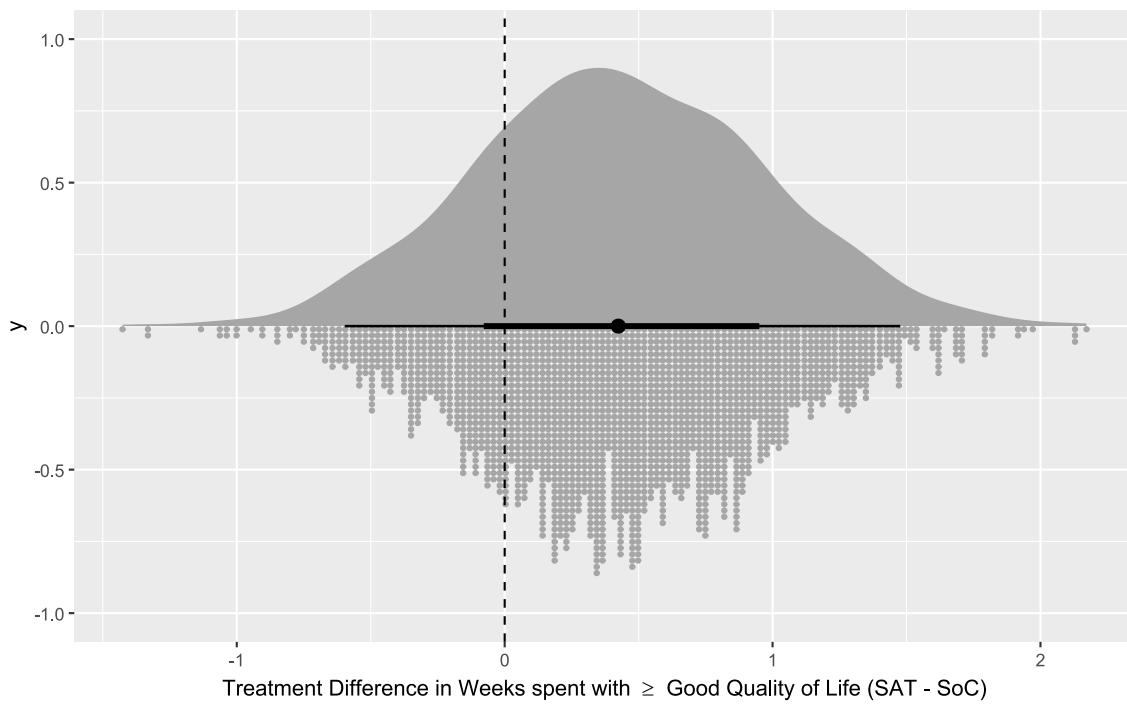
In the large dataset the true difference between $tx == 1$ and $tx == 0$ is 1.5 weeks.

```
# marginalized difference in weeks for  $\geq$  good QoL
posterior <- sop_df |>
  filter(state == "1" | state == "2" | state == "3") |>
  group_by(tx, draw) |>
  summarise(mean_weeks = sum(sop)) |>
  pivot_wider(names_from = tx, values_from = mean_weeks, names_prefix = "tx_") |>
  mutate(difference = tx_1 - tx_0) |>
  ggplot(aes(x = difference)) +
  ggdist::stat_dots(side = "bottom") +
  ggdist::stat_halfeye() +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(
    x = bquote(paste("Treatment Difference in Weeks spent with ", ""  $\geq$  "", " Good
```

```
Quality of Life (SAT - SoC"))  
)
```

``summarise()`` has grouped output by 'tx'. You can override using the ``.groups`` argument.

```
posterior
```



```
ggsave(here::here("img", "primary-estimand-posterior.svg"), width = 8, height = 5)
```